4. Gemini Integration

The Gemini SDK/API is connected from the backend. The API key is kept in an environment variable rather than public frontend code.

Tabulation

Module	Function
Frontend	Collect user preferences
Backend	Validate input and call Gemini
AI	Generate plan
Output	Display structured result

Coding / Technical Example

```
from google import genai

client = genai.Client(api_key="YOUR_API_KEY")

response = client.models.generate_content(
    model="gemini-3.8-flash",
    contents=prompt
)

print(response.text)
```

Picture / Diagram

```
FRONTEND → FLASK BACKEND → GEMINI API → RESPONSE → RESULT
```

5. Response Formatting

The generated text can be displayed under sections such as Weekly Schedule, Activity, Warm-up, Cool-down and Recovery.

Tabulation

Module	Function
Frontend	Collect user preferences
Backend	Validate input and call Gemini
AI	Generate plan
Output	Display structured result

Coding / Technical Example

```
from google import genai

client = genai.Client(api_key="YOUR_API_KEY")

response = client.models.generate_content(
    model="gemini-3.8-flash",
    contents=prompt
)

print(response.text)
```

Picture / Diagram

```
FRONTEND → FLASK BACKEND → GEMINI API → RESPONSE → RESULT
```

6. Example Implementation

The following example shows the basic server-side generation flow.

Tabulation

Module	Function
Frontend	Collect user preferences
Backend	Validate input and call Gemini
AI	Generate plan
Output	Display structured result

Coding / Technical Example

```
from google import genai

client = genai.Client(api_key="YOUR_API_KEY")

response = client.models.generate_content(
    model="gemini-3.8-flash",
    contents=prompt
)

print(response.text)
```

Picture / Diagram

```
FRONTEND → FLASK BACKEND → GEMINI API → RESPONSE → RESULT
```